
# Weichi Zhao

**AI Infrastructure Engineer | LLM Serving, MaaS, Kubernetes**

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Target roles: AI infrastructure, LLM serving platform, cloud-native model platform, AI gateway, heterogeneous GPU scheduling.

## Profile

AI infrastructure engineer with production experience across LLM serving platforms, OpenAI-compatible MaaS, model gateways, offline batch inference, Kubernetes operators, and heterogeneous accelerator environments. I focus on turning model services into reliable platform capabilities: clear control/data-plane boundaries, observable routing, safe rollout, online/offline traffic coordination, and failure recovery that is verified end to end rather than only at the Pod level.

## Core Skills

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| **LLM Serving** | SGLang, vLLM, OpenAI-compatible APIs, streaming, embeddings, Responses API compatibility, PD disaggregation, KV-cache-aware routing. |
| **Platform** | Kubernetes, operators/controllers, CRDs, Helm, Kustomize, rollout safety, service discovery, GPU task lifecycle management. |
| **Gateway** | Higress, Envoy/WASM concepts, LiteLLM, authentication, model mapping, token-level rate limiting, routing, observability. |
| **Engineering** | Go, Python, Java, Linux, Prometheus metrics, MySQL/SQLite/Redis, distributed systems debugging, performance testing. |

## Experience

### Zhejiang Lab, Supercomputing Research Center
Aug 2021 - Present

Engineer, AI infrastructure and MaaS platform

- Built and operated LLM serving platform capabilities across CPU ACK and GPU ACK clusters, covering private model deployments through Higress, public model services through LiteLLM, SGLang Router based model routing, IGD/IGDSA inference orchestration, and BatchFlow offline inference.
- Owned production troubleshooting and hardening for LLM serving paths, including Router worker registration convergence, Service/Endpoint readiness, PD prefill/decode topology, CrashLoop and long-connection failures, gateway header/model mapping issues, and multi-cluster rollout validation.
- Designed control-plane ideas for MaaS-Controller and optimizer loops: collect LiteLLM, Router, IGDSA, BatchFlow, and gateway metrics; adjust public model limits and offline concurrency with observation windows, rollback criteria, and online-SLA-first rules.
- Worked on earlier cloud-native AI/HPC platform projects, including Kubernetes adaptation on domestic OS and accelerators, device plugin customization for accelerator availability, PyTorch/model adaptation, and distributed training framework exploration.

## Selected AI Infrastructure Projects

### BatchFlow - OpenAI-compatible offline inference platform
**LLM batch inference**

Asynchronous JSONL batch execution for large-scale, non-real-time model workloads.

- Designed and implemented Batch API compatible capabilities: file upload, batch creation, status query, cancellation, result/error files, streaming file processing, request-level error isolation, and backend concurrency control.
- Refactored BatchFlow toward multi-replica operation with Kubernetes leader election and sharded workers, improved header forwarding, added upload limits, automatic stale-file cleanup, Kustomize-based upgrades, monitoring views, and failure-tolerant finalization logic.
- Supported production data-distillation workloads: over 2.76M requests and about 23.9B tokens processed during a holiday run; later supported daily processing around 10M requests with success rate above 99% under controlled concurrency.
- Validated a BatchFlow -> SGLang Router -> multi-KLX backend path for Qwen3-235B-A22B-Thinking-2507 at about 300K requests, about 9.75B tokens, and about 99.9% success rate; confirmed BatchFlow and Router overhead stayed below 5% CPU/memory in the test profile.

### SGLang Router, IGD, and production serving reliability
**Routing and orchestration**

Reliable worker registration, routing, and recovery for multi-worker and PD-disaggregated inference.

- Improved Router worker synchronization by reconciling desired workers against live `/workers` state and requeueing on missing, extra, or failed registrations, reducing hidden partial-registration failures.

- Enhanced IGD/ISI behavior around Router startup and readiness by preferring Pod IPs during early registration, enabling `publishNotReadyAddresses`, preserving Service finalizers, and adding Prometheus Service generation for Router metrics ports.
- Developed and validated ISI-level auto-restart capabilities with explicit opt-in labels, restart counters, failure fingerprints, retry budget, cleanup of failed resources, and forward compatibility for existing workloads.
- Led production root-cause analysis for failures such as ARG_MAX overflow from accumulated service-link environment variables, Router not managed after Worker Ready, stale Service selectors, and long-connection timeout behavior.

**MaaS gateway and control-plane integration** — **Higress, LiteLLM, metrics**

Unified model access, routing, rate limiting, and observability for private and public model services.

- Participated in the platform architecture that separates Higress private deployment entry, LiteLLM public model entry, SGLang Router model routing, BatchFlow offline execution, and MaaS-Controller control-plane coordination.
- Investigated and validated AI gateway capabilities including token-level rate limits, model mapping, route synchronization, metrics collection, OpenAI-compatible protocol handling, and compatibility boundaries for embeddings and Responses-style APIs.
- Resolved user-facing integration issues by tracing requests across gateway, router, worker, service discovery, and model protocol layers instead of relying only on top-level deployment health.

## Earlier Projects

**Arizona State University, SNAC Lab** — Jan 2020 - Aug 2020

Virtual platform developer and research assistant

- Developed modules for THOTH Lab, an open virtual lab platform for education and research, including SDN/OpenFlow lab modules and AWS instance control through Boto3-backed REST APIs.
- Implemented an Android prototype for an IR-CP-ABE based network security research system using Java.

**Arizona State University, CIDSE** — Aug 2019 - Dec 2020

Graduate service assistant

- Assisted CSE205 Object-Oriented Programming, CSE412 Database Management, CSE434 Computer Networks, and CSE464 Software QA and Testing.

## Education

**Arizona State University** — May 2019 - Dec 2020

Master of Computer Science, GPA 3.97/4.00

**Arizona State University** — Aug 2015 - May 2019

Bachelor of Science in Computer Science, GPA 3.40/4.00

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